RUHR UNIVERSITÄT BOCHUM

Bachelor/Master Thesis Topic

The robustness of NL-based Web GUI tests

Motivation and Background:

Script-based web GUI testing frameworks (Playwright, Cypress) dominate the testing landscape, yet they suffer from a fundamental fragility: their reliance on DOM locators causes tests to break when applications evolve, even when underlying functionality remains unchanged [1]. This maintenance burden is well documented and represents a significant cost in software development cycles [2]. Natural Language (NL)-based testing has emerged as a promising alternative to address this fragility. In a recent thesis by Junior Gogang Kengne [3], a novel runtime-based framework was developed that automatically converts script-based GUI tests into NL-based tests through a record, translate, and validate process. The framework captures test actions, converts them into high-level NL instructions using LLMs, and then validates the resulting test outcome. Simultaneously, Leon Adamietz's recent thesis introduced the Reproducible Locator Break Dataset (RLBD) [4], which contains hundreds of reproducible locator breaks extracted from real open-source repositories. This dataset serves as an ideal testbed for verifying whether NL-based testing genuinely offers the claimed robustness improvements by evaluating whether NL tests can resist locator breaks that would cause script-based tests to break.

This thesis has two connected goals: (1) re-evaluating the existing framework [3] with greater methodological rigor, and (2) systematically assessing whether NL-based tests are robust to survive locator breaks by executing converted tests against the RLBD reproduction breaks.

Student Task and Responsibilities:

- Refine and extend the existing framework [3] to make it a maintainable, configurable, and publishable tool, then re-evaluate it across a broader set of web applications using well-established metrics in the literature.
- Create a test harness that translates reproduction break tests from the RLBD into executable NL-based test evaluations. For each reproducible break, run the corresponding NL-based test (if successfully converted) and verify if it passes despite the locator break that would cause script-based tests to fail.
- Assess the robustness of the NL-based test by measuring the percentage of NL-converted tests that survive locator breaks. Create a thorough empirical evaluation.

Deliverables:

- An enhanced and publishable-ready tool of the NL-based test conversion framework with improved quality and comprehensive documentation.
- A rigorous empirical evaluation report of how many converted NL-based tests survive locator breaks compared to script-based versions.
- A report of the corner and reasons for failures on the NL conversion and/or for the failure in handling the locator break scenario.

Pre-Requisites: (Programming Languages, OS, Skills, Papers, etc)

- Experience with Python, JavaScript, and TypeScript.
- Familiarity with Playwright and Cypress frameworks, and a solid understanding of web technologies (DOM, CSS selectors, HTML attributes).
- Understanding of LLM fundamentals, NLP concepts, and prompt engineering.
- Experience with Docker for using the RLDB.

[1] Nass, Michel, Emil Alégroth, and Robert Feldt. "Why many challenges with GUI test automation (will) remain." Information and Software Technology 138 (2021): 106625.

[2] Hammoudi, Mouna, Gregg Rothermel, and Paolo Tonella. "Why do Record/Replay Tests of Web Applications Break?" 2016 IEEE International Conference on Software Testing, Verification and Validation (ICST). IEEE, 2016.

[3] Kengne, Junior Gogang. "Automated Migration of Script-based Web GUI Tests to NL-based Web GUI Tests." Bachelor's Thesis, Ruhr University Bochum, 2026.

[4] Adamietz, Leon. "Identification and Reproduction of Locator Breaks in Evolving Web Applications." Bachelor's Thesis, Ruhr University Bochum, 2026.

Contacts

Thiago Santos de Moura, M.Sc. (`sq-teaching@rub.de`)
Software Quality group, Faculty of Computer Science, Ruhr University of Bochum